

CertiChain: A Blockchain-Based Secure Certificate Verification System

PROJECT SYNOPSIS

OF PROJECT-I (PROJICB781)

BACHELOR OF TECHNOLOGY

in

Computer Science & Engineering
(Cyber Security)

SUBMITTED BY

ASHWANI RAJ (13031722047)

DEEPEN PAUL (13031722048)

NIKHIL KUMAR (13031722049)

NAVONIL AMIN (13031722050)



**Department of Computer Science & Engineering
(Cyber Security)**

Techno Main Salt Lake.

Kolkata -700091.



*Department of Information Technology
Techno Main Salt Lake
EM-4/1, Salt Lake City, Kolkata-700091*

Supervisor's Verification

This is verified that the synopsis entitled “ **CertiChain: A Blockchain-Based Secure Certificate Verification System** ” is a bonafide part of the final year project work being carried out by **ASHWANI RAJ , DEEPEN PAUL , NIKHIL KUMAR , NAVONIL AMIN** , students of B.Tech in **Computer Science & Engineering(Cyber Security)**, under my supervision.

The content presented in the synopsis is in accordance with the academic standards and guidelines of the department.

Full Signature of the Candidates (with date)

1. _____

2. _____

3. _____

4. _____

(Signature of the Supervisor)

(Signature of the Supervisor)

(Signature of the Head of the Department)

Contents

1	Abstract	4
2	Literature Review	4
3	Problem Definition	6
4	Facilities Required	7
5	Methodology / Planning of Work	8
6	Conclusion	11

1 Abstract

With the increasing reliance on digital credentials in the academic, professional, and skill-based sectors, ensuring the authenticity of certificates has become a critical security concern. Traditional verification methods are often slow, manual, and prone to forgery, leading to significant risks for institutions, employers, and individuals. This project proposes a blockchain-based certificate verification system that provides immutable proof of authenticity while enabling instant, cost-free verification without intermediaries. By leveraging decentralized ledger technology, the system ensures that certificate records are tamper-proof and transparently verifiable, enhancing trust between issuers, holders, and verifiers.

The methodology involves generating a unique cryptographic hash for each issued certificate, storing this hash along with relevant metadata on a blockchain network, and providing a web-based interface for real-time verification. Certificate files are stored in decentralized storage (IPFS), and only their hashes are recorded on-chain to maintain privacy while ensuring integrity. The proposed system implements a streamlined verification process that can be integrated into institutional workflows, reducing administrative overhead and mitigating fraudulent claims.

The verification framework incorporates smart contract logic for record creation, retrieval, and validation, combined with a user friendly frontend for issuers and verifiers. The research aims to deliver a secure, scalable, and cost-effective solution capable of operating on public blockchain test networks, offering a pathway for large-scale deployment in real-world educational and corporate environments.

2 Literature Review

The verification of educational and professional certificates has long faced issues such as forgery, delays, and reliance on centralized authorities. While digitization has improved access and speed, centralized systems remain vulnerable to single points of failure and security breaches.

Blockchain offers a decentralized, immutable ledger for storing certificate data or cryptographic hashes, ensuring tamper-proof and transparent [8]. Frameworks like **Blockcerts** by MIT demonstrate the feasibility of blockchain-anchored credentials.

Hash-based models, such as **OpenCerts** and **OpenAttestation** by **GovTech Singapore**, store certificates off-chain (e.g., IPFS) while recording only their hashes on-chain, reducing costs and preserving privacy. In contrast, centralized systems like **India's National Academic Depository (NAD)** improve accessibility but lack blockchain's decentralization, prompting calls for integration.[7]

International initiatives like the **European Blockchain Services Infrastructure (EBSI)** adopt **W3C Verifiable Credentials** to standardize cross-border verification and enable selective disclosure, balancing trust, security, and privacy.

Pros of Existing Methods:

- **Security and Tamper-Resistance** – Immutable blockchain records prevent unauthorized alterations.[3]
- **Privacy Preservation** – Off-chain storage with cryptographic hashing protects sensitive information.[9]
- **Global Interoperability** – Standards like W3C VC ensure cross-platform compatibility.[9]
- **Cost Efficiency** – Hash-based models reduce blockchain storage expenses.[8]
- **Instant Verification** – Eliminates delays caused by manual processing.[3]

Cons of Existing Methods:

- **Scalability Challenges** – High transaction costs on public blockchains without optimization.
- **Revocation Complexity** – Difficult to modify or revoke records due to immutability.[9]
- **Technical Barriers** – Requires blockchain infrastructure and technical literacy.[3]
- **Regulatory Gaps** – Lack of uniform legal frameworks across jurisdictions.[8]
- **Adoption Hurdles** – Institutions may be slow to transition from centralized models.
[9]

3 Problem Definition

The authenticity of educational and professional certificates is critical for academic admissions, employment, and professional licensing. However, the current verification process faces several challenges:

Challenges in Current Verification Process:

- **1. Forgery and Fraud** – Fake certificates are often produced using advanced editing tools, making manual detection difficult.
- **2. Time-Consuming Verification** – Traditional verification requires physical document submission or email confirmation from issuing institutions, often taking days or weeks.
- **3. Centralized Dependence** – Most verification systems rely on a single institutional database, creating a single point of failure vulnerable to hacking, tampering, or outages.
- **4. Lack of Cross-Institution Trust** – Certificates issued in one country or institution are often not easily verifiable by another without lengthy procedures.
- **5. Cost and Resource Overhead** – Institutions must maintain dedicated staff and infrastructure for verification, adding operational costs.

Requirement of Blockchain Technology:

Given these limitations, there is a pressing need for a secure, decentralized, and instant verification system that ensures the integrity, authenticity, and transparency of certificates without relying on intermediaries.[3]

Blockchain technology provides an immutable, tamper-proof ledger where certificate data or its cryptographic hash can be stored, enabling instant and trustless verification from anywhere in the world.[9] This approach significantly reduces verification time, eliminates forgery risks, and enhances global trust.

4 Facilities Required

To design, develop, and test the proposed Blockchain-Based Certificate Verification System at zero or minimal cost, the following facilities are required:

- **1. Hardware Requirements:**

- **Minimum** - Dual-core processor, 4 GB RAM, 500 GB HDD, Windows/Linux/macOS OS.
- **Recommended** - Quad-core processor, 8 GB+ RAM, SSD storage for faster blockchain compilation and deployment.
- **Internet Connectivity** – Stable broadband connection for blockchain network access, smart contract deployment, and hosting.
- **External Storage (Optional)** – For backups and storing large media files (if certificate images are high resolution).

- **2. Software Requirements (Free / Open Source):**

- **Frontend Development** – HTML5, CSS3, JavaScript (Vanilla or React.js), Bootstrap/Tailwind CSS for responsive UI.[4]
- **Backend Development** – Node.js + Express.js (or Firebase Functions for serverless).[4]
- **Blockchain Development Tools** – Remix IDE (online Solidity editor for writing and deploying smart contracts)[8].
MetaMask (wallet for interacting with blockchain testnets).[8]
Ganache / Hardhat / Truffle (for local blockchain development).[8]
Infura / Alchemy (free API to connect to Ethereum testnets like Sepolia).
- **Database** – MongoDB Atlas (free tier) or Firebase Firestore (NoSQL) for storing off-chain data.[2]
- **File Storage (Decentralized)** – IPFS via web3.storage or Pinata (free tiers).
- **Version Control** – Git and GitHub for code repository and collaboration.
- **Hosting** – GitHub Pages / Netlify / Vercel for frontend.
Render / Railway for backend APIs.

5 Methodology / Planning of Work

The development of the Blockchain-Based Certificate Verification System will follow a structured and phased approach to ensure accuracy, security, and efficiency.

- **Phase 1: Requirement Analysis**

Identify Stakeholders: Issuers (universities, institutions), Holders (students), Verifiers (employers, other institutions).

- **Gather Functional Requirements:** Certificate issuance process.[3]
Blockchain integration for proof storage.[3]
Instant verification mechanism.[3]
Revocation and update support.[3]
- **Non-Functional Requirements:** Security, scalability, usability, and privacy compliance.

- **Phase 2: System Design**

- **Architecture Planning:** Decentralized blockchain layer for immutable proof storage.
Off-chain database for metadata and certificate details.[7]
IPFS for decentralized file storage.[7]
- **Module Design:**
 - 1. Issuer Module** – Allows institutions to create and issue certificates.[1]
 - 2. Verification Module** – Enables verifiers to upload a certificate and check authenticity.[1]
 - 3. User Interface Module** – Web application for interaction between stakeholders.[1]

- **Phase 3: Blockchain Smart Contract Development**

Develop smart contracts in Solidity for storing certificate hashes and metadata.

- **Implement functions:**
 - issueCertificate() – Record hash and metadata on blockchain.[3]
 - verifyCertificate() – Compare uploaded hash with stored hash.[3]
 - revokeCertificate() – Mark invalid certificates.[3]
 - Deploy to Ethereum Sepolia testnet using Remix IDE and MetaMask.[3]

- **Phase 4: Frontend Backend Development**

- **Frontend:** HTML/CSS/JavaScript (React.js optional) for responsive UI.[5]
MetaMask integration for blockchain transactions.[5]
- **Backend:** Node.js with Express.js or Firebase Functions.
REST APIs for certificate data retrieval.[6]
- **Database Integration:** MongoDB Atlas or Firebase Firestore for off-chain metadata. IPFS integration for storing actual certificate files.[2]

- **Phase 5: Testing Validation**

- **Functional Testing:** Check all features (issuance, verification, revocation).
- **Performance Testing:** Measure transaction time, system scalability, and IPFS file retrieval speed.
- **Cross-Platform Testing:** Test on mobile, tablet, and desktop browsers.

- **Phase 6: Deployment**

- **Frontend Hosting:** GitHub Pages, Netlify, or Vercel.
- **Backend Hosting:** Render or Railway free tier.
- **Blockchain:** Sepolia testnet for initial deployment, with migration plan to mainnet if needed.[3]

- **Phase 7: Documentation and Training**

- Prepare User Manual for issuers and verifiers.
- Maintain Technical Documentation (API docs, contract details).[6]
- Conduct demo/training sessions for intended users.

- **Phase 8: Maintenance and Future Enhancements**

- Implement Decentralized Identity (DID) integration for secure authentication.
- Add Mobile App for quick verification using QR codes.

Proposed Workflow Diagram

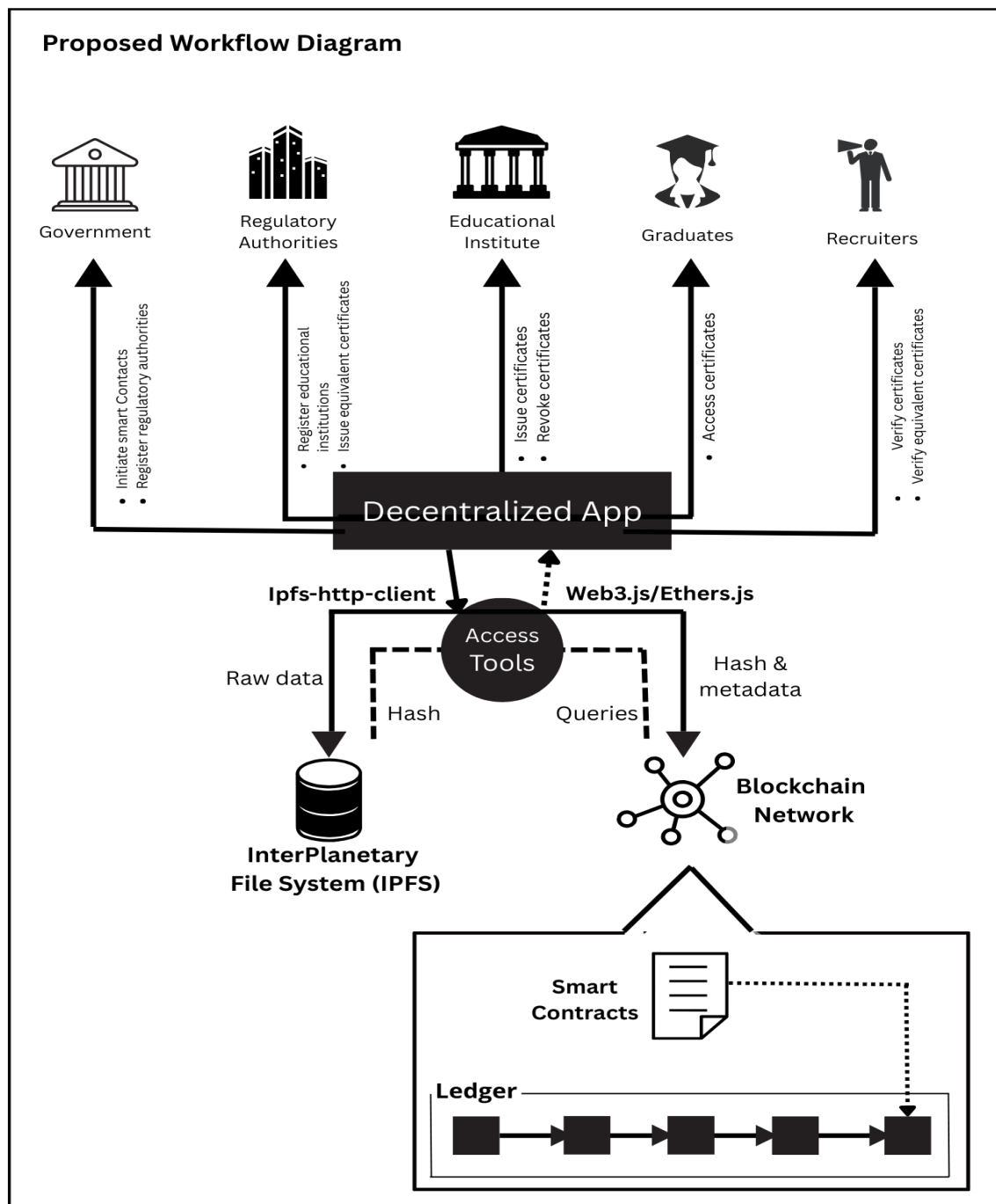


Fig 1.1 Workflow Diagram of a Blockchain based Certificate Verification System

6 Conclusion

The Blockchain-Based Certificate Verification System provides a secure, transparent, and tamper-proof solution to the growing problem of forged and manipulated academic credentials. By leveraging blockchain technology, certificate data becomes immutable and can be verified instantly without intermediaries, ensuring trust between institutions, employers, and individuals.[1]

The system also integrates decentralized storage like IPFS to store original certificates securely, removing dependency on a single server and reducing the risk of data loss. Institutions benefit from reduced administrative workload, employers save time with real-time verification, and students gain a lifelong, portable proof of their achievements.

Beyond solving current verification challenges, this system has the potential to standardize credential authentication globally, paving the way for a universal trust network for academic and professional records. With future enhancements such as smart contracts and integration with national databases, it can become a global benchmark for secure certificate verification.

References

- [1] Andrew W. Appel. Verification of a cryptographic primitive: Sha-256. *ACM Transactions on Programming Languages and Systems (TOPLAS)*, 37(2):1–31, 2015.
- [2] Kyle Banker, Douglas Garrett, Peter Bakkum, and Shaun Verch. *MongoDB in Action: Covers MongoDB Version 3.0*. Simon and Schuster, 2016.
- [3] Chris Dannen. Solidity programming. In *Introducing Ethereum and Solidity: Foundations of Cryptocurrency and Blockchain Programming for Beginners*, pages 69–88. Springer, 2017.
- [4] Danny Goodman. *Dynamic HTML: The Definitive Reference: A Comprehensive Resource for HTML, CSS, DOM & JavaScript*. O’Reilly Media, Inc., 2002.
- [5] Harish A. Jartaghar, Girish Rao Salanke, Ashok Kumar A.R., Sharvani G.S., and Shivakumar Dalali. “React apps with server-side rendering: Next.js,” *Journal of Telecommunication, Electronic and Computer Engineering (JTEC)*, 14(4):25–29, 2022.
- [6] JMF. Java. API, 2013.
- [7] Neha Kewate, Amruta Raut, Mohit Dubekar, Yuvraj Raut, and Ankush Patil. “A review on AWS-cloud computing technology,” *International Journal for Research in Applied Science and Engineering Technology*, 10(1):258–263, 2022.
- [8] Samudya Nanayakkara, M.N.N. Rodrigo, Shamal Perera, Geeganage Thilini Weerasuriya, and Amer A. Hijazi. “A methodology for selection of a blockchain platform to develop an enterprise system,” *Journal of Industrial Information Integration*, 23:100215, 2021.
- [9] Michael Nofer, Peter Gomber, Oliver Hinz, and Dirk Schiereck. “Blockchain,” *Business & Information Systems Engineering*, 59(3):183–187, 2017.
- [10] Antonio Quiña-Mera, Pablo Fernandez, José María García, and Antonio Ruiz-Cortés. “GraphQL: A systematic mapping study,” *ACM Computing Surveys*, 55(10):1–35, 2023.